

Here's a MATLAB script that computes the regression lines for two-dimensional random variables using the given statistical measures derived from a joint probability mass function (jpmf) or joint probability density function (jpdf).

Formulas for Regression Lines

The regression lines of Y on X and X on Y are given by:

$$E[Y|X] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (X - \mu_X)$$

$$E[X|Y] = \mu_X + \rho \frac{\sigma_X}{\sigma_Y} (Y - \mu_Y)$$

where:

- μ_X, μ_Y are the means of X and Y .
- σ_X, σ_Y are the standard deviations of X and Y .
- ρ is the correlation coefficient.

% MATLAB script to compute regression lines for two-dimensional random variables

clc; clear; close all;

% Define the joint probability mass function (jpmf) or joint probability density function (jpdf)

% Example: Joint PMF for discrete random variables

X = [1, 2, 3]; % Possible values of X

Y = [1, 2, 3]; % Possible values of Y

**jpmf = [0.1, 0.2, 0.1;
0.2, 0.1, 0.1;
0.1, 0.1, 0.1]; % Joint probability mass function**

% Compute marginal distributions

px = sum(jpmf, 2); % P(X)

py = sum(jpmf, 1); % P(Y)

% Compute means

mu_X = sum(X .* px');

mu_Y = sum(Y .* py);

% Compute standard deviations

sigma_X = sqrt(sum(((X - mu_X).^2) .* px));

sigma_Y = sqrt(sum(((Y - mu_Y).^2) .* py));

% Compute covariance

E_XY = sum(sum((X' * Y) .* jpmf)); % E[XY]

cov_XY = E_XY - mu_X * mu_Y;

% Compute correlation coefficient

rho = cov_XY / (sigma_X * sigma_Y);

% Regression lines

% E[Y | X] = mu_Y + rho * (sigma_Y / sigma_X) * (X - mu_X)

EY_X = mu_Y + rho * (sigma_Y / sigma_X) * (X - mu_X);

% E[X | Y] = mu_X + rho * (sigma_X / sigma_Y) * (Y - mu_Y)

EX_Y = mu_X + rho * (sigma_X / sigma_Y) * (Y - mu_Y);

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% Display results
fprintf('Mean of X: %.4f\n', mu_X);
fprintf('Mean of Y: %.4f\n', mu_Y);
fprintf('Standard deviation of X: %.4f\n', sigma_X);
fprintf('Standard deviation of Y: %.4f\n', sigma_Y);
fprintf('Correlation coefficient (rho): %.4f\n', rho);

% Plot regression lines
figure;
scatter(X, EY_X, 'ro', 'filled'); hold on;
plot(X, EY_X, 'r-', 'LineWidth', 2);
scatter(EX_Y, Y, 'bo', 'filled');
plot(EX_Y, Y, 'b-', 'LineWidth', 2);
legend('E[Y|X]', 'Regression Line Y on X', 'E[X|Y]', 'Regression Line X on Y');
xlabel('X and Y values');
ylabel('Conditional Expectations');
title('Regression Lines');
grid on;

```

Explanation of the MATLAB Code

1. Define Joint PMF (or JPDF for continuous variables)
 - The joint probability values are stored in `jpmf`.
 - `X` and `Y` represent the possible values of the random variables.
2. Compute Marginal Distributions
 - The marginal probabilities are calculated by summing over rows/columns.
3. Calculate Mean (μ_X, μ_Y)
 - Using the marginal probabilities.
4. Compute Standard Deviations (σ_X, σ_Y)
 - Based on variance formulas.
5. Compute Covariance and Correlation Coefficient (ρ)
 - Using the expected value of XY .
6. Compute Regression Equations
 - Using the formulas for $E[Y|X]$ and $E[X|Y]$.
7. Plot the Regression Lines
 - Scatter plot and regression lines are plotted for visual analysis.

Practice Problems

1. Given is the joint probabilities of X and Y.

Y \ X	0	1	2
0	0.02	0.08	0.10
1	0.05	0.20	0.25
2	0.03	0.12	0.15

2. The joint probability mass function pmf of X and Y is

Y \ X	0	1	2
0	0.1	0.04	0.02
1	0.08	0.20	0.06
2	0.06	0.14	0.30

3. In an automobile plant two tasks are performed by robots. The first entails welding two joints; the second, tightening three bolts. Let X denote the number of defective welds and Y the number of improperly tightened bolts produced per car. Since X and Y are each, discrete, (X, Y) is a two-dimensional discrete random variable. Past data indicates that the joint density for (X, Y) is as shown in the Table.

Y \ X	0	1	2	3
0	0.840	0.030	0.020	0.010
1	0.060	0.010	0.008	0.002
2	0.010	0.005	0.004	0.001

4. The two common types of errors made by programmers are syntax errors and errors in logic. For a simple language such as BASIC the number of such errors is usually small. Let X denote the number of syntax errors and Y the number of errors in logic made on the first run of a BASIC program. Assume that the joint density for (X, Y) is as shown in the Table.

Y \ X	0	1	2	3
0	0.400	0.100	0.020	0.005
1	0.300	0.040	0.010	0.004
2	0.040	0.010	0.009	0.003
3	0.009	0.008	0.007	0.003
4	0.008	0.007	0.005	0.002
5	0.005	0.002	0.002	0.001

5. Two random variables X and Y have the joint density, $f(x, y) = 2 - x - y$, $0 \leq x \leq 1$, $0 \leq y \leq 1$; 0, otherwise
6. X and Y are two random variables having joint density function $f(x, y) = (1/8)(6 - x - y)$ for $0 < x < 2$ and $2 < y < 4$.
7. The joint probability function of two random variables X and Y is given by $f(x, y) = c(2x+y)$, where x and y can assume all integers such that $0 \leq x \leq 2$ and $0 \leq y \leq 3$ and $f(x, y)=0$ otherwise
8. Two random variables X and Y have the joint density $f(x, y) = x y$, $0 \leq x \leq 1$, $0 \leq y \leq 2$
9. The joint pdf of two random variables is given by $f(x, y) = 6 \exp[-2x-3y]$ for $x \geq 0$, $y \geq 0$.